

The empirical recurrence for OEIS A257421, recovered from its tail

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8 September 2026

Abstract

OEIS A257421 records “Empirical recurrence of order 88” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 400$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 95$. Its last coefficient is $c_{88} = 32 \neq 0$, so no further tail satisfies anything shorter; any order-88 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

2020 Mathematics Subject Classification. 05A15, 11B37, 15A18.

1 A conjecture that is not written down

OEIS A257421 is “Number of $(n+2) \times (3+2)$ 0..1 arrays with every 3×3 subblock diagonal sum plus antidiagonal minimum nondecreasing horizontally, vertically and ne-to-sw antidiagonally.”. It has offset 1 and begins

10684, 27244, 117324, 449022, 1290910, 3917190, 11883298, 33766938, ...

The entry states:

Empirical recurrence of order 88 (see link above)

As of the “Last modified” line on the live entry (Jul 23 2025 15:23:11, revision 6) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 400$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 400$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 88: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 88.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 800$ exact terms from $n = 7$ on, it returns order 88 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{88} c_i a(n-i),$$

c_1	7	c_{23}	73331	c_{45}	33948	c_{67}	−3470
c_2	−17	c_{24}	−121043	c_{46}	−39454	c_{68}	7455
c_3	17	c_{25}	91154	c_{47}	22581	c_{69}	−3206
c_4	−8	c_{26}	6525	c_{48}	24099	c_{70}	−1917
c_5	5	c_{27}	−20349	c_{49}	−23312	c_{71}	3375
c_6	22	c_{28}	−16807	c_{50}	−27633	c_{72}	−2769
c_7	−178	c_{29}	−16213	c_{51}	38544	c_{73}	−190
c_8	452	c_{30}	85747	c_{52}	−30630	c_{74}	2348
c_9	−546	c_{31}	−88822	c_{53}	11049	c_{75}	−1712
c_{10}	508	c_{32}	21110	c_{54}	22548	c_{76}	248
c_{11}	−683	c_{33}	−35776	c_{55}	−17927	c_{77}	860
c_{12}	871	c_{34}	112786	c_{56}	959	c_{78}	−996
c_{13}	−410	c_{35}	−95087	c_{57}	6128	c_{79}	232
c_{14}	−890	c_{36}	20369	c_{58}	−15539	c_{80}	360
c_{15}	2578	c_{37}	−5275	c_{59}	3829	c_{81}	−432
c_{16}	−5076	c_{38}	35589	c_{60}	10167	c_{82}	152
c_{17}	8303	c_{39}	−48493	c_{61}	−10541	c_{83}	168
c_{18}	−12703	c_{40}	44140	c_{62}	11703	c_{84}	−216
c_{19}	18765	c_{41}	−18187	c_{63}	−2046	c_{85}	112
c_{20}	−26027	c_{42}	−24188	c_{64}	−9085	c_{86}	0
c_{21}	31409	c_{43}	39855	c_{65}	6669	c_{87}	−64
c_{22}	−39350	c_{44}	−31934	c_{66}	−1907	c_{88}	32

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 95$, and it is the recurrence OEIS A257421 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{88} - c_1x^{88-1} - \dots - c_{88}$. Its constant term is $-c_{88} = -32$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k\lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 88 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 88, so $\deg q = 88$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 88, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 88, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is 32.

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-88 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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